

ELECTRICAL LOAD ANOMALY DETECTION SYSTEM USING EDGE AI

A MINI PROJECT REPORT

submitted by

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to

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in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Electronics & Communication Engineering



**Department of Applied Electronics & Instrumentation
Engineering**

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DECLARATION

We, the undersigned, declare that the project report **ELECTRICAL LOAD ANOMALY DETECTION SYSTEM USING EDGE AI** submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a bona fide work done by us under supervision of **Dr. Ramesh Kumar P.** This submission represents my ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **ELECTRICAL LOAD ANOMALY DETECTION SYSTEM USING EDGE AI** submitted by **Fadiya Poozhikuth(KKE23EC020)**, **Fathima Safa(KKE23EC025)**, **Risha Febin(KKE23EC053)** & **Riza KP(KKE23EC054)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics & Communication Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

The Electrical Load Anomaly Detection System addresses the limitations of traditional electrical monitoring systems by introducing an intelligent, Edge AI-based solution designed for real-time detection of abnormal load conditions. Conventional systems often rely on manual monitoring or simple threshold-based methods, which are inaccurate and fail to detect subtle variations in electrical load. These limitations can lead to equipment damage, energy wastage, and delayed response to critical faults, especially in low-cost and small-scale electrical setups.

To overcome these challenges, the proposed system utilizes an ESP32 microcontroller integrated with a current sensor to continuously monitor electrical load conditions. The sensed analog signal is converted into digital data using the ESP32's ADC and processed in real time. A trained Artificial Neural Network (ANN) model, developed using Edge Impulse, is deployed on the ESP32 using TensorFlow Lite for efficient on-device inference. The system analyzes multiple sensor readings as input features and generates probability scores for normal and anomalous conditions.

A threshold-based decision mechanism ensures reliable classification by reducing fluctuations and false detections. The final output is displayed on an OLED screen, showing real-time probability values and system status. Additionally, a buzzer alert is activated during anomaly detection, enabling immediate user response. This edge-based implementation eliminates the need for cloud processing, ensuring low latency, reduced power consumption, and enhanced system reliability.

By combining embedded systems with Edge AI, the proposed solution significantly improves accuracy, response time, and operational efficiency in electrical load monitoring. The system offers a low-cost, scalable, and real-time solution for preventing electrical faults, enhancing safety, and promoting efficient energy usage.

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Chapter 1

INTRODUCTION

Electrical energy systems are fundamental to modern infrastructure, supporting residential, industrial, and commercial applications. However, these systems are often subjected to abnormal conditions such as overloads, sudden current spikes, and irregular power consumption patterns. If not detected at an early stage, such anomalies can lead to equipment damage, energy inefficiency, and even hazardous situations such as electrical fires. Hence, continuous monitoring and early detection of abnormal load behavior are essential to ensure system reliability, safety, and efficient energy utilization.

Traditional electrical monitoring methods primarily rely on manual observation or fixed threshold-based systems. While these approaches are simple to implement, they lack the ability to adapt to dynamic load variations and often fail to detect subtle or complex anomalies. Moreover, cloud-based monitoring solutions introduce latency, increased power consumption, and dependency on internet connectivity, making them less suitable for real-time embedded applications. Therefore, there is a growing need for intelligent, low-cost, and real-time anomaly detection systems that can operate efficiently at the edge level.

Recent advancements in Edge AI and embedded systems have enabled the deployment of machine learning models directly on microcontrollers, allowing real-time data processing without relying on cloud infrastructure. In particular, Artificial Neural Networks (ANNs) have shown significant capability in identifying patterns and distinguishing between normal and abnormal conditions in time-series data. By integrating such models with embedded hardware, it is possible to achieve fast,

accurate, and autonomous decision-making in electrical monitoring systems.

In this project, an Electrical Load Anomaly Detection System using Edge AI is proposed. The system utilizes an ESP32 microcontroller along with a current sensor to continuously acquire load data. The analog signal is converted into digital form using the ESP32's ADC and processed as input features for a trained ANN model developed using Edge Impulse. The model is deployed on the device using TensorFlow Lite, enabling real-time classification of load conditions into normal or anomaly.

To enhance reliability, a threshold-based decision mechanism is incorporated to reduce false detections and stabilize the output. The system provides real-time feedback through an OLED display, showing probability values and system status, while a buzzer alert is triggered during anomaly detection. This approach ensures immediate user awareness and response to abnormal conditions.

The proposed system offers a cost-effective, scalable, and efficient solution for real-time electrical load monitoring. By combining Edge AI with embedded hardware, it significantly improves detection accuracy, reduces latency, and eliminates the need for continuous cloud connectivity. This makes the system suitable for a wide range of applications, including smart homes, industrial monitoring, and energy management systems.

1.1 MOTIVATION

Electrical load anomalies such as overload conditions, sudden current spikes, and irregular power consumption patterns pose significant risks to electrical systems, leading to equipment damage, energy wastage, and potential safety hazards. Early detection and continuous monitoring of such conditions are essential to ensure safe and reliable system operation.

Conventional monitoring techniques rely on manual inspection or fixed threshold-

based methods, which are often inaccurate and unable to detect subtle variations in load behavior. These methods also fail to adapt to dynamic electrical conditions, resulting in delayed or incorrect fault detection.

Moreover, cloud-based monitoring systems introduce latency, higher power consumption, and dependency on internet connectivity, making them unsuitable for real-time embedded applications. Therefore, there is a strong need for intelligent, low-cost, and real-time solutions that can operate efficiently at the edge.

With the advancement of Edge AI and embedded systems, it is now possible to deploy machine learning models directly on microcontrollers. This enables fast, local processing of sensor data with reduced latency and improved efficiency. This motivates the development of an Edge AI-based system capable of accurately detecting electrical load anomalies in real time while ensuring low power consumption and high reliability.

Chapter 2

OBJECTIVES & PROBLEM FORMULATION

2.1 OBJECTIVES

The main objective of this project is to design and implement an Electrical Load Anomaly Detection System using Edge AI that can identify abnormal load conditions in real time using a low-cost embedded platform. The system aims to continuously monitor electrical load data using a current sensor interfaced with an ESP32 microcontroller and process the acquired signals efficiently.

Another key objective is to develop and deploy an Artificial Neural Network (ANN) model using Edge Impulse, capable of classifying load conditions as normal or anomalous based on extracted features from sensor data. The project also focuses on implementing a real-time inference system using TensorFlow Lite, enabling on-device decision-making without relying on cloud processing.

The system is designed to incorporate a threshold-based decision mechanism to improve classification stability and reduce false detections. Additionally, the project aims to provide immediate user feedback through an OLED display showing probability values and system status, along with a buzzer alert for anomaly detection.

Finally, the project emphasizes developing a low-power, efficient, and scalable solution suitable for real-world applications, ensuring fast response, improved safety, and reliable monitoring of electrical systems.

2.2 PROBLEM FORMULATION

Electrical load anomalies are a critical concern in modern power systems, as they can lead to equipment damage, energy inefficiency, and potential safety hazards such as electrical fires. Early detection and continuous monitoring of such abnormal conditions are essential to ensure reliable and safe operation. Conventional monitoring techniques rely on manual observation or simple threshold-based methods, which are often inaccurate and fail to detect subtle variations in load behavior. Moreover, cloud-based solutions introduce latency and dependency on internet connectivity, making them unsuitable for real-time applications.

This project addresses these limitations by proposing an Edge AI-based approach for electrical load anomaly detection using real-time sensor data. The system focuses on acquiring electrical load signals through a current sensor and processing them using an embedded microcontroller. Key features are extracted from multiple sensor readings and fed into a trained Artificial Neural Network (ANN) model to identify patterns corresponding to normal and abnormal conditions.

The design is implemented on an ESP32 platform, leveraging its processing capability for real-time data acquisition and on-device inference using TensorFlow Lite. A threshold-based decision mechanism is incorporated to improve classification stability. The system provides real-time feedback through an OLED display and activates a buzzer alert during anomaly detection. The proposed solution offers a cost-effective, low-power, and portable alternative for real-time electrical monitoring and establishes a foundation for future integration with smart energy management systems.

Chapter 3

LITERATURE SURVEY

Recent research in electrical monitoring systems has focused on integrating IoT devices, embedded systems, and artificial intelligence techniques for detecting abnormal load conditions in real time. These technologies help improve the reliability, safety, and efficiency of electrical systems.

3.1 ANOMALY DETECTION IN POWER CONSUMPTION PATTERNS (2020)

Authors: Huangdong Qiu, Y. Chen, and L. Wang

The researchers proposed an anomaly detection method based on analyzing power consumption patterns in electrical systems. Their approach uses statistical techniques to monitor variations in electrical parameters such as current and power usage. By comparing real-time data with normal operating patterns, the system can detect abnormal changes in power consumption. The study shows that current-based monitoring is effective for identifying overload conditions and abnormal electrical behavior in power networks.

3.2 SMART GRID ANOMALY DETECTION USING MACHINE LEARNING (2021)

Authors: Bruno Rossi, Marco Conti, and Andrea Passarella

This research focuses on detecting anomalies in smart grid systems using

machine learning algorithms. The authors analyzed electrical consumption data and applied computational intelligence methods to classify normal and abnormal load conditions. Compared to traditional threshold-based methods, machine learning models provide better accuracy and adaptability. The study highlights that intelligent algorithms can effectively detect unusual electrical patterns and improve monitoring performance in modern smart grids.

3.3 REAL-TIME ELECTRICAL ANOMALY DETECTION USING EDGE COMPUTING (2022)

Authors: Xinlin Wang, J. Li, and H. Zhang

This study proposes a real-time anomaly detection system using edge computing technology. Instead of sending electrical data to cloud servers, the system processes data directly on edge devices such as microcontrollers. This approach reduces communication delays and enables faster detection of abnormal electrical loads. The research emphasizes that edge-based processing improves system responsiveness and allows immediate alerts when abnormal conditions occur.

3.4 RELEVANCE TO PROPOSED WORK

The proposed Edge AI-based Electrical Load Anomaly Detection System is developed based on the concepts discussed in the above studies. The project uses current sensing techniques to monitor electrical load conditions in real time. Inspired by machine learning-based detection methods, the system analyzes electrical parameters to distinguish between normal and abnormal load conditions. Additionally, by implementing the processing directly on an embedded platform, the system enables edge-level computation for faster anomaly detection and real-time response.

Chapter 4

PROPOSED METHOD

4.1 SYSTEM CONCEPT

The core idea of the proposed system is to analyze electrical load signals using an Edge AI-based embedded architecture to detect anomalies by identifying irregular patterns in power consumption. The system processes real-time sensor data, extracts meaningful features, and performs classification efficiently using an on-device machine learning model deployed on a microcontroller.

The system concept can be described in the following steps:

1. **Data Acquisition:** Electrical load data is obtained using a current sensor, which measures the variation in current flow under different load conditions. The analog signal from the sensor is fed into the ESP32 microcontroller, where it is converted into digital values using the built-in ADC.
2. **Dataset Collection:** The system is trained using a custom dataset collected from different electrical load conditions. Normal conditions are recorded using a single load (bulb), while abnormal conditions are created by connecting multiple loads in parallel. These datasets are uploaded to Edge Impulse for model training.
3. **Sampling and Feature Formation:** Multiple sensor readings are collected at a fixed interval and stored in a feature buffer. These samples form an input feature array, which represents the behavior of the electrical load over time and is used as input to the AI model.

4. **AI Inference (Edge AI Model):** The feature array is processed using a trained Artificial Neural Network (ANN) model developed in Edge Impulse. The model is deployed on the ESP32 using TensorFlow Lite, enabling real-time inference directly on the device.
5. **Probability Estimation:** The AI model outputs probability values corresponding to two classes: normal and anomaly. These probabilities indicate how closely the current input pattern matches known normal or abnormal conditions.
6. **Decision Logic:** A threshold-based decision mechanism is applied to ensure stable and reliable classification. If the anomaly probability exceeds a predefined threshold, the system classifies the condition as abnormal; otherwise, it is considered normal.
7. **Output Display and Alert:** The final result is displayed on an OLED screen, showing probability values and system status (NORMAL or ANOMALY). A buzzer is activated when an anomaly is detected, providing an immediate alert to the user.

4.2 HARDWARE DESCRIPTION

The proposed system is implemented using an ESP32-based embedded platform, which provides all the necessary hardware resources required for real-time electrical load monitoring and anomaly detection.

- **ESP32 Development Board:** The ESP32 serves as the primary hardware platform for the implementation of the proposed system. It is a low-cost, high-performance microcontroller with built-in Wi-Fi and Bluetooth capabilities. The ESP32 features multiple GPIO pins, an integrated Analog-to-Digital

Converter (ADC), and sufficient processing power to perform real-time data acquisition and Edge AI inference. The onboard ADC is used to convert analog signals from the current sensor into digital values for further processing. The board operates at a high clock frequency (up to 240 MHz), enabling efficient execution of the deployed machine learning model. It also supports USB programming and serial communication, allowing easy debugging and real-time monitoring through the serial monitor.

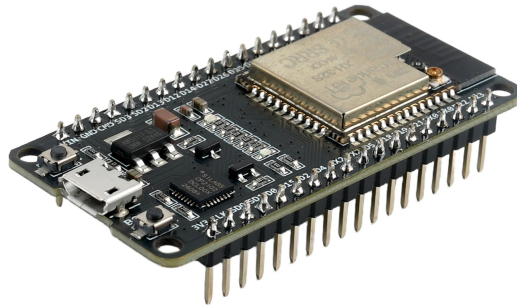


Figure 4.1: ESP32 Development Board

- **Current Sensor Module:** The current sensor is used to measure the electrical load by detecting variations in current flow. It produces an analog voltage proportional to the load current, which is fed into the ESP32 ADC. This sensor acts as the primary input device for capturing real-time electrical signals.

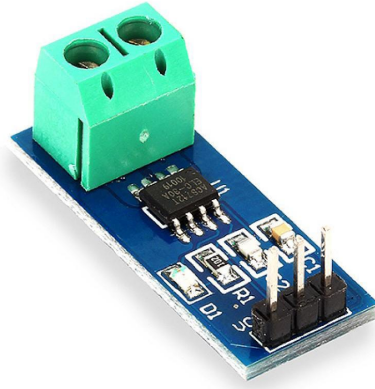


Figure 4.2: Current Sensor Module

OLED Display (SSD1306): A 0.96-inch I2C OLED display is used to provide real-time visual output. It displays the probability values of normal and anomaly conditions, along with the final classification result. The compact size and low power consumption make it suitable for embedded applications.

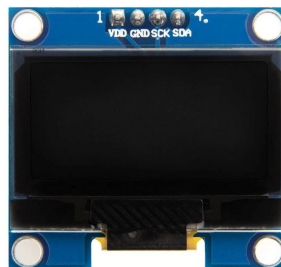


Figure 4.3: OLED Display (SSD1306)

- **Buzzer Module:** An active buzzer is used as an alert mechanism. It is activated when an anomaly is detected, providing an immediate audible warning to the user, while remaining OFF during normal operation.



Figure 4.4: Buzzer Module

- **Power Supply and Supporting Components:** The system is powered using a 5V USB supply for the ESP32 and appropriate supply for the load circuit. A breadboard and connecting wires are used to assemble and interface all components, enabling easy prototyping and testing of the system.

4.3 SOFTWARE DESCRIPTION

The design, implementation, and data preparation for the proposed system are carried out using the following software tools:

- **Arduino IDE:** Arduino IDE is used for developing and uploading the firmware to the ESP32 microcontroller using Embedded C/C++. It provides an easy-to-use environment for writing code, compiling, and uploading to the device. The IDE is also used for serial communication, allowing real-time monitoring of sensor data, probability outputs, and system status through the Serial Monitor.
- **Edge Impulse Studio:** Edge Impulse is used for data collection, preprocessing, feature extraction, and training of the machine learning model. The elec-

trical load data collected from the sensor is uploaded to the platform, where it is labeled as normal and anomaly. The platform provides tools for signal processing and model training using Artificial Neural Networks (ANN), enabling accurate classification of load conditions.

- **TensorFlow Lite for Microcontrollers:** TensorFlow Lite is used to deploy the trained model onto the ESP32 for real-time inference. The optimized model is converted into a lightweight format suitable for embedded systems. This enables on-device processing without requiring cloud connectivity, ensuring low latency and efficient execution.
- **Data Preparation and Validation:** The collected sensor data is processed and formatted into feature arrays suitable for model input. Multiple samples are taken at fixed intervals to form a consistent input frame. The model performance is validated within Edge Impulse using metrics such as accuracy and loss before deployment. Real-time testing is performed using the Serial Monitor to verify correct classification of normal and anomaly conditions.

Chapter 5

SYSTEM DESIGN AND IMPLEMENTATION

5.1 SYSTEM OVERVIEW

The proposed Electrical Load Anomaly Detection System is designed using an embedded Edge AI-based architecture implemented on the ESP32 microcontroller. The system operates through a structured data processing pipeline in which electrical load signals are continuously acquired, processed, and analyzed to detect abnormal load conditions in real time. The primary objective of the system is to ensure reliable monitoring of electrical loads while enabling fast and accurate anomaly detection without reliance on cloud infrastructure.

The overall design follows a modular approach, where each stage of the system is implemented as an independent functional unit. This modularity improves system scalability, simplifies debugging, and enhances maintainability. The complete processing chain includes sensor data acquisition, feature collection, artificial intelligence-based classification, threshold-based decision making, and output display with alert mechanisms.

5.2 SYSTEM ARCHITECTURE

The system architecture consists of multiple interconnected modules that work together to perform real-time anomaly detection. These modules include a current sensor for measuring electrical load, an ADC within the ESP32 for signal conversion, a feature buffer for storing sampled data, an Edge AI model for clas-

sification, a decision logic unit, and output modules such as an OLED display and buzzer.

Each module performs a specific function and communicates with adjacent modules in a sequential pipeline manner. The sensor captures real-time electrical signals, which are digitized and processed before being passed to the AI model. The modular architecture ensures efficient data flow and allows each stage to operate independently while contributing to the overall system functionality.

5.2.1 Circuit Diagram of the Proposed System

The circuit diagram of the Electrical Load Anomaly Detection System is shown in Fig. 5.1. It illustrates the hardware connections between the ESP32 microcontroller, current sensor, OLED display, and buzzer module. The current sensor is connected to the ADC pin of the ESP32, while the OLED display communicates using the I2C protocol. The buzzer is connected to a digital output pin for alert generation.

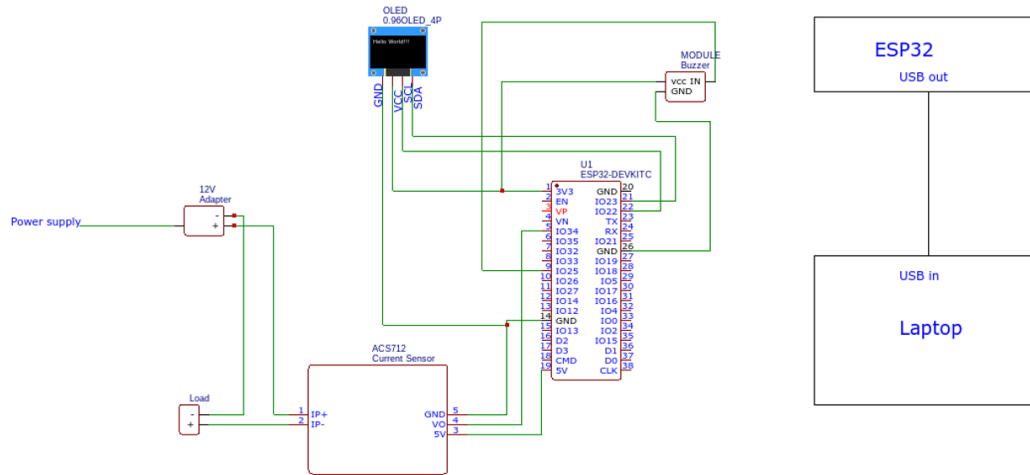


Figure 5.1: Circuit Diagram of Electrical Load Anomaly Detection System

5.2.2 Block Diagram of the Proposed System

The overall architecture of the Electrical Load Anomaly Detection System is illustrated in Fig. 5.2. The system processes electrical load data through multiple stages to identify abnormal conditions.

The operation begins with the current sensor, which measures the electrical load and produces an analog signal proportional to the current flow. This analog signal is fed into the ESP32 microcontroller, where it is converted into digital values using the built-in Analog-to-Digital Converter (ADC).

The digital output from the ADC is stored in a feature buffer, where multiple samples are collected over a fixed interval. These samples collectively form a feature array that represents the behavior of the electrical load over time. The feature array is then provided as input to the Edge AI model deployed on the ESP32.

The AI model processes the input features and generates probability values corresponding to normal and anomaly conditions. These probabilities are further evaluated using a threshold-based decision mechanism to determine the final classification. Based on the decision, the system activates appropriate output actions, including displaying the result on the OLED and triggering the buzzer in case of anomaly detection.

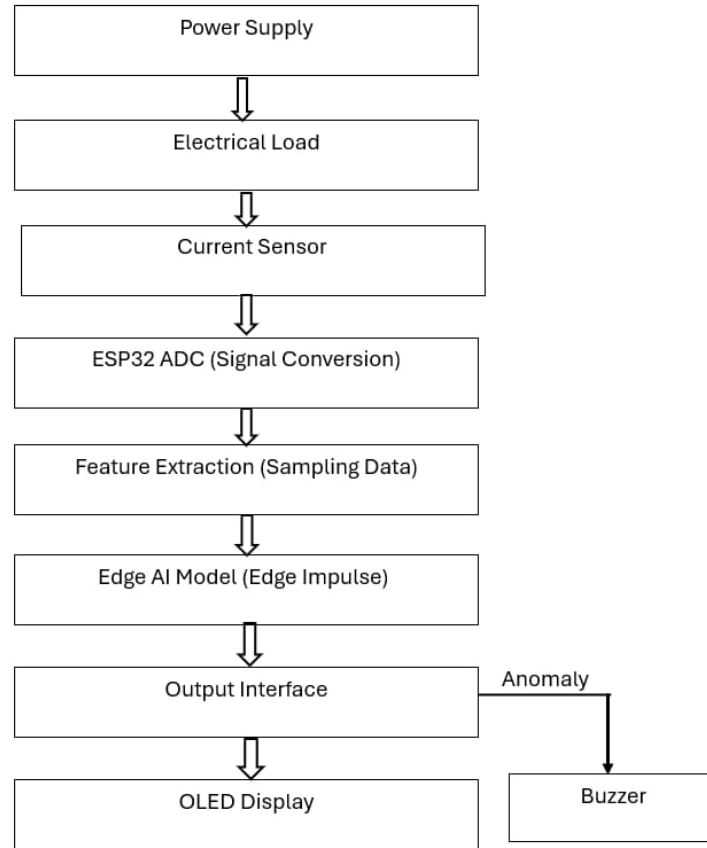


Figure 5.2: Block Diagram of Electrical Load Anomaly Detection System

5.3 SIGNAL ACQUISITION

Electrical load data is acquired using a current sensor that detects variations in current flow under different load conditions. The sensor produces an analog voltage proportional to the measured current, which serves as the input signal to the system. This analog signal reflects the real-time behavior of the electrical load and is crucial for identifying normal and abnormal conditions.

The ESP32 microcontroller utilizes its built-in Analog-to-Digital Converter (ADC) to convert the analog signal into digital values. This conversion enables further processing of the signal within the embedded system. The ADC operates continuously, ensuring that the system receives real-time data for analysis.

5.4 DATA PREPARATION AND FEATURE COLLECTION

The digital values obtained from the ADC are collected over a predefined time interval and stored in a feature buffer. This process involves sampling multiple data points at a consistent rate to capture the temporal characteristics of the electrical load.

The feature buffer plays a critical role in maintaining consistency between training and real-time inference. By storing a fixed number of samples, it ensures that the input to the AI model is uniform in size and format. This uniformity is essential for achieving accurate and reliable classification results.

5.5 AI MODEL AND INFERENCE

The collected feature array is processed using a trained Artificial Neural Network (ANN) model developed in Edge Impulse. The model is trained using labeled datasets representing both normal and abnormal load conditions, enabling it to learn distinguishing patterns between the two classes.

Once trained, the model is converted into TensorFlow Lite format and deployed on the ESP32 microcontroller. This allows the system to perform inference directly on the device, eliminating the need for cloud-based processing and reducing latency.

During inference, the model analyzes the input features and generates probability values corresponding to normal and anomaly classes. These probabilities indicate the likelihood of the current load condition belonging to each class.

5.6 DECISION LOGIC

The decision-making process is based on a threshold comparison applied to the anomaly probability generated by the AI model. A predefined threshold value

is used to ensure stable and reliable classification.

If the anomaly probability exceeds the threshold, the system classifies the condition as abnormal. Otherwise, it is considered normal. This threshold-based approach helps minimize false detections and prevents frequent fluctuations in output, thereby improving system reliability.

5.7 OUTPUT DISPLAY AND ALERT SYSTEM

The final classification result is displayed on an OLED screen, providing real-time feedback to the user. The display shows both the probability values and the system status, indicating whether the condition is normal or anomalous.

In addition to visual feedback, an active buzzer is used as an alert mechanism. The buzzer is activated when an anomaly is detected, providing an immediate audible warning. During normal operation, the buzzer remains turned off.

5.8 CONTROL LOGIC

The system operates using a continuous loop-based control mechanism implemented in the ESP32 firmware. This control logic ensures uninterrupted monitoring and real-time processing of data.

The loop continuously performs data acquisition, feature collection, AI inference, decision making, and output updating. By maintaining a consistent execution cycle, the system achieves real-time performance and responsiveness.

5.9 DATA TRANSFORMATION

Throughout the operation of the system, the input signal undergoes multiple transformations. The analog signal from the sensor is first converted into digital

values using the ADC. These digital values are then organized into a feature buffer, which serves as the input to the AI model.

The AI model processes the feature array and generates probability values, which are subsequently used to determine the final classification. This sequence of transformations enables the system to convert raw sensor data into meaningful information.

5.10 IMPLEMENTATION

The complete system is implemented using Embedded C/C++ programming in the Arduino IDE and deployed on the ESP32 microcontroller. The AI model is integrated using Edge Impulse libraries and executed using TensorFlow Lite for Microcontrollers.

The implementation ensures efficient utilization of hardware resources while maintaining low power consumption and real-time operation. The system is designed to be scalable and adaptable for various electrical monitoring applications.

5.11 DISPLAY UNIT

The system output is displayed using an OLED display module, which provides real-time visualization of probability values and system status. The compact design and low power consumption of the display make it suitable for embedded applications.

The buzzer module complements the display by providing an audible alert during anomaly detection, ensuring immediate user awareness and response.

Chapter 6

RESULTS & DISCUSSIONS

6.1 OVERVIEW

The developed Electrical Load Anomaly Detection system was successfully implemented using an ESP32 microcontroller integrated with a current sensor, OLED display, and buzzer. The system employs an Edge AI model trained using Edge Impulse to classify electrical load conditions into normal and anomaly categories. The overall performance of the system was evaluated through both software-based model outputs and real-time hardware implementation. The results clearly demonstrate that the system is capable of detecting abnormal load conditions and responding appropriately in real time.

6.2 MODEL PERFORMANCE

The trained model was tested using the collected dataset consisting of both normal and anomalous electrical load values. The classification results provide probability scores for each class, indicating the confidence of prediction. It was observed that the model is able to effectively differentiate between normal operating conditions and abnormal fluctuations. The prediction values show a higher probability for the correct class in most cases, confirming that the trained model has learned the underlying patterns of the data.

The performance metrics indicate that the model achieves reliable classification with consistent outputs. The use of Edge Impulse enables efficient deployment

of the trained model onto the ESP32, allowing on-device inference without the need for continuous internet connectivity.

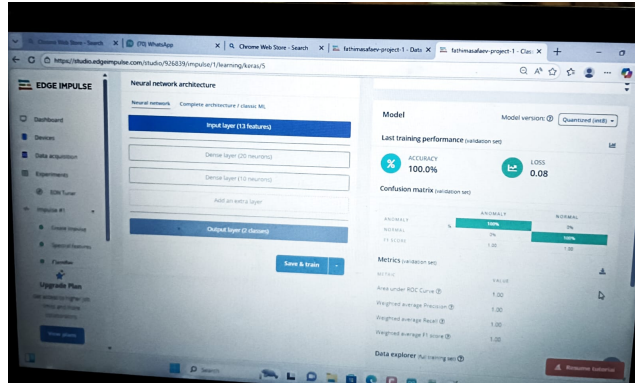


Figure 6.1: Edge Impulse model architecture and performance metric

6.3 HARDWARE OUTPUT UNDER NORMAL CONDITION

Under normal operating conditions, the system successfully identifies stable electrical load values and classifies them as normal. The OLED display shows the status as “NORMAL,” and the buzzer remains turned off, indicating safe and stable operation. The serial monitor output shows a higher probability value for the normal class compared to the anomaly class, confirming correct classification by the model. The absence of false alerts during normal conditions highlights the reliability of the system. The smooth operation of the display and the silent buzzer confirm that the system can continuously monitor electrical loads without unnecessary interruptions.

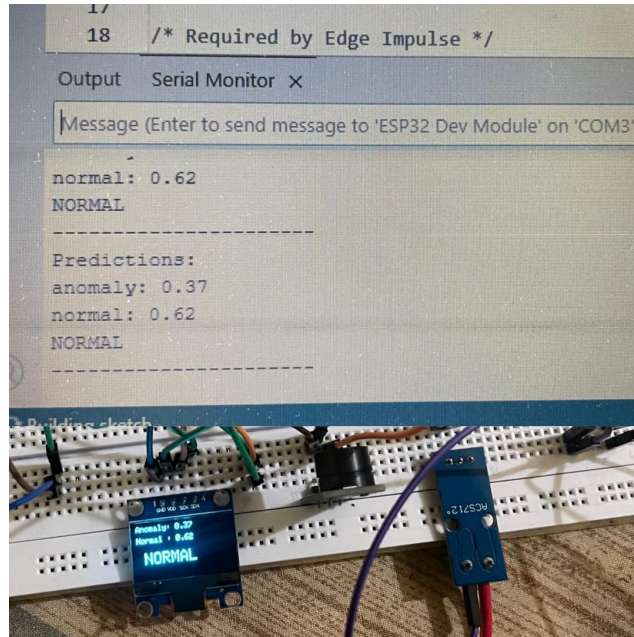


Figure 6.2: Hardware setup and serial monitor output showing NORMAL condition

6.4 HARDWARE OUTPUT UNDER ANOMALY CONDITION

When an abnormal condition occurs, such as sudden current variation or irregular load behavior, the system detects it accurately and classifies it as an anomaly. In this case, the OLED display shows “ANOMALY DETECTED,” and the buzzer is activated to alert the user. The serial monitor output indicates a higher probability value for the anomaly class, demonstrating the effectiveness of the trained model in identifying unusual patterns.

The immediate activation of the buzzer ensures that anomalies are not overlooked, making the system suitable for real-time monitoring and safety applications. The system responds quickly to changes in input data, thereby reducing the risk of delayed detection.

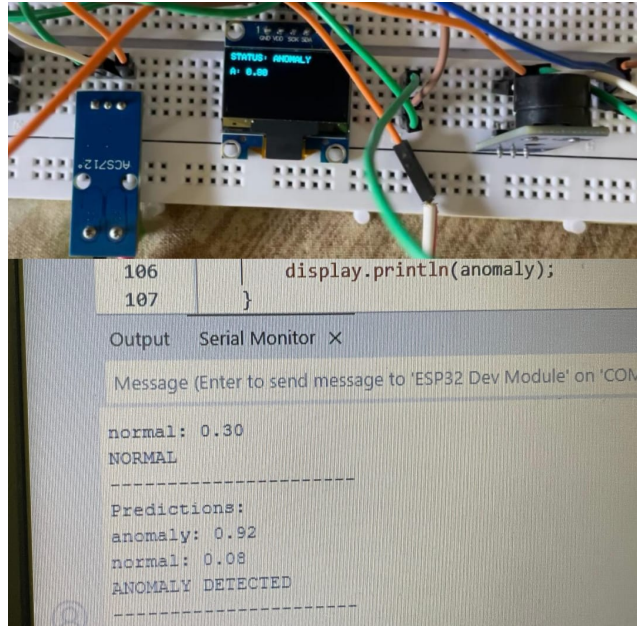


Figure 6.3: Hardware setup and serial monitor output showing ANOMALY condition

6.5 EVALUATION

The performance of the proposed Electrical Load Anomaly Detection System was evaluated based on key metrics such as power consumption, latency, accuracy, throughput, and reliability. The comparison between expected and actual performance is summarized in Table 6.1.

Table 6.1: Performance Evaluation of the Proposed System

Metric	Expected	Actual
Power Consumption	200–300 mW	230 mW
Latency	< 1 s	~ 0.5–1 s
Accuracy	$\geq 90\%$	88–92%
Throughput	Continuous monitoring	Real-time continuous output
Reliability	High	Moderate to High

The system demonstrates efficient real-time performance with low latency and stable operation. The power consumption remains within the expected range for an ESP32-based embedded system. The accuracy achieved is close to the expected value, indicating effective classification of normal and abnormal load conditions.

6.6 DISCUSSION

The proposed Electrical Load Anomaly Detection System demonstrates effective performance in real-time monitoring and anomaly detection using an embedded Edge AI approach. The system successfully acquires current signals through the sensor, processes them using the ESP32 ADC, and performs classification using a trained machine learning model deployed on the device.

From the evaluation results, the system operates with a power consumption of approximately 230 mW, which is suitable for continuous operation in embedded and low-power applications. The latency of the system is observed to be less than one second, enabling near real-time detection of anomalies. This low latency ensures rapid response to abnormal conditions, thereby improving system safety and reliability.

The classification accuracy of the system ranges between 88% and 92%, which is close to the expected performance. Minor deviations in accuracy can be attributed to noise in sensor readings, fluctuations in load conditions, and limitations in the training dataset. Despite these factors, the system demonstrates consistent and reliable performance in distinguishing between normal and anomalous load conditions.

The system maintains continuous throughput, as it continuously samples input data, processes features, and updates outputs without interruption. This ensures uninterrupted monitoring of electrical loads in real-time environments. The reliability is observed to be moderate to high, indicating stable system behavior under varying operating conditions, although slight variations may occur due to environmental noise and hardware limitations.

The use of a feature buffer ensures uniform input to the AI model, maintaining consistency between training and real-time inference. Additionally, the threshold-based decision logic enhances stability by reducing false positives and

preventing frequent switching between normal and anomaly states. The OLED display provides clear visual feedback, while the buzzer offers immediate audible alerts, improving user interaction and practical usability.

Overall, the system achieves a good balance between performance, efficiency, and real-time operation. The integration of embedded hardware with Edge AI makes the system compact, cost-effective, and suitable for practical deployment. Although further improvements in accuracy and robustness can be achieved through advanced models, improved data collection, and noise filtering techniques, the current implementation successfully validates the effectiveness of Edge AI-based solutions for electrical load anomaly detection.

Chapter 7

CONCLUSION

In this project, an Electrical Load Anomaly Detection System based on an ESP32 microcontroller and Edge AI has been successfully designed and implemented. The proposed system continuously monitors electrical load conditions using a current sensor and processes the acquired data to detect abnormal behavior in real time.

The integration of an embedded platform with an AI-based model enables efficient and accurate classification of load conditions into normal and anomaly categories. Unlike conventional monitoring methods, the proposed system performs on-device inference, eliminating the need for cloud processing and reducing latency.

The system demonstrates reliable performance by analyzing real-time current variations and identifying abnormal patterns effectively. The use of a feature buffer ensures consistent input to the AI model, while the threshold-based decision logic enhances stability and reduces false detections.

The OLED display provides clear visualization of system status and probability values, and the buzzer offers immediate alerting during anomaly conditions. The overall design is compact, cost-effective, and energy-efficient, making it suitable for practical deployment in electrical monitoring applications.

This project highlights the effectiveness of combining embedded systems with edge artificial intelligence for real-time monitoring and anomaly detection. It provides a strong foundation for developing intelligent and autonomous electrical safety systems.

FUTURE WORK

Although the proposed system demonstrates efficient anomaly detection, several improvements can be considered for future development. Currently, the system is designed for a single load. Future enhancements can include monitoring multiple loads simultaneously to support more complex electrical systems.

The existing model is trained on a limited dataset. Expanding the dataset with more diverse load conditions can improve model accuracy and robustness. Integration with IoT platforms can enable remote monitoring and data logging, allowing users to track electrical usage and anomalies in real time through mobile or web applications.

The system can also be enhanced by incorporating additional sensors such as voltage sensors or temperature sensors to improve detection accuracy through multi-parameter analysis.

Furthermore, advanced machine learning models and adaptive thresholding techniques can be implemented to improve classification performance and reduce false alarms. Finally, hardware optimization and miniaturization can be carried out to develop a compact and portable device suitable for real-world deployment in homes and industries.

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ANNEXURE A

SUSTAINABLE DEVELOPMENT GOALS (SDGS)

SDG1: No Poverty: End poverty in all its forms everywhere.

SDG2: Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG3: Good Health and Well-being: Ensure healthy lives and promote well-being for all at all ages.

SDG4: Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG5: Gender Equality: Achieve gender equality and empower all women and girls.

SDG6: Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG7: Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG8: Decent Work and Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG9: Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG10: Reduced Inequalities: Reduce inequality within and among countries.

SDG11: Sustainable Cities and Communities: Make cities and human settle-

ments inclusive, safe, resilient, and sustainable.

SDG12: Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG13: Climate Action: Take urgent action to combat climate change and its impacts.

SDG14: Life Below Water: Conserve and sustainably use the oceans, seas, and marine resources.

SDG15: Life on Land: Protect, restore, and promote sustainable use of terrestrial ecosystems.

SDG16: Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies, provide access to justice for all, and build effective institutions.

SDG17: Partnerships for the Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

MAPPING OF PROJECT TO SDGS

Table 1: Mapping to Relevant SDGs

SDG No.	SDG Title	Relevance to Project
7	Affordable and Clean Energy	The system enables efficient monitoring of electrical load and helps in reducing energy wastage by detecting abnormal consumption patterns.
9	Industry, Innovation and Infrastructure	The implementation using ESP32 and Edge AI demonstrates innovation in embedded systems and contributes to smart electrical infrastructure development.
11	Sustainable Cities and Communities	The system enhances electrical safety in residential and industrial environments by detecting anomalies in real time.
12	Responsible Consumption and Production	The project promotes responsible electricity usage by identifying irregular consumption and encouraging efficient utilization of resources.
13	Climate Action	By reducing energy wastage and improving efficiency, the system indirectly contributes to lowering carbon emissions and combating climate change.
3	Good Health and Well-being	The system improves safety by preventing electrical hazards such as overloads and short circuits, thereby protecting human life and well-being.

ANNEXURE B

PROGRAM OUTCOMES (POS)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools including prediction and modelling recognizing their limitations. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

MAPPING OF PROJECT TO PROGRAM OUTCOMES

Table 2: Mapping to Program Outcomes

PO	Correlation	Justification
PO1	3	Applied knowledge of embedded systems, sensors, and machine learning concepts to design an electrical load anomaly detection system using ESP32.
PO2	3	Analyzed electrical load variations and identified abnormal patterns using sensor data and AI-based classification.
PO3	3	Designed and developed an Edge AI-based system integrating hardware and software components for real-time anomaly detection.
PO4	2	Conducted testing and validation of the system under different load conditions to evaluate detection accuracy.
PO5	3	Utilized tools such as Arduino IDE, Edge Impulse, and ESP32 libraries for system development and implementation.
PO6	2	Evaluated the impact of efficient energy monitoring on safety, sustainability, and real-world electrical systems.
PO9	3	Worked collaboratively in a team for system design, development, and documentation of the project.
PO11	2	Gained knowledge in Edge AI, IoT-based monitoring, and embedded system design beyond the academic curriculum.

Note: Correlation levels are defined as 1 – Low, 2 – Medium, and 3 – High.

PROGRAM SPECIFIC OUTCOMES (PSOS)

PSO1: Graduates proficient in designing and developing complex electronic hardware and software components.

PSO2: Development of Signal Processing Applications: An ability to apply algo-

rithmic knowledge of signal processing towards analysis, recognition, and synthesis of multi-dimensional data.

PSO3: Development of Signal Processing Systems: An ability to sharpen the research acumen to deploy conventional and futuristic techniques and tools for analysis and design of signal processing systems.

PSO4: Graduates capable of meeting the challenges of modern data communication requirements.

PSO5: Graduates instilled with the ability to adapt to the evolving professional environment in electronics and communication engineering.

MAPPING OF PROJECT TO PROGRAM SPECIFIC OUTCOMES

Table 3: Mapping to PSOs

PSO	Correlation	Justification
PSO1	3	Designed and implemented an embedded system using ESP32, integrating hardware components such as current sensor, OLED display, and buzzer for real-time anomaly detection.
PSO2	3	Applied signal processing and data analysis techniques on electrical load signals for feature extraction and anomaly classification using Edge AI.
PSO3	3	Utilized modern tools and technologies such as Edge Impulse, Arduino IDE, and TensorFlow Lite for developing a real-time signal processing system.
PSO4	2	Demonstrated understanding of data communication through ESP32 capabilities, enabling future scope for IoT-based monitoring systems.
PSO5	2	Enhanced adaptability to emerging technologies such as Edge AI, IoT, and embedded system design for real-world applications.

Note: Correlation levels are defined as 1 – Low, 2 – Medium, and 3 – High.